

UNIVERSITY OF BAHRAIN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER 2025 -2026

TEST 1 (B) - Key

MATHS. 211

DATE: 13 - 10 - 2025

TIME: 11:00 PM - 12:00 PM

EXAM INSTRUCTIONS:

- 1) Number of questions is 3 and number of pages is 5.
- 2) Show all your work, except for question one.

NAME :	I.D. NUMBER:	SECTION:	SERIAL NUMBER:

<i>Q.1</i>	<i>Q.2</i>	<i>Q.3</i>	<i>OUT OF</i>
<i>8</i>	<i>5</i>	<i>7</i>	

GOOD LUCK

QUESTION 1 (1 marks for each question)

Choose the correct answer, then fill in the blank in the following table:

Questions	1	2	3	4	5	6	7	8
Answers	<i>a</i>	<i>b</i>	<i>a</i>	<i>c</i>	<i>b</i>	<i>a</i>	<i>d</i>	<i>d</i>

1) If $A = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 3 & -2 \\ 1 & 3 & 4 \end{pmatrix}$, then the reduced row echelon form of A is

(a) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 4 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

2) If A, B and C are matrices such that A has size 3×4 , B has size 3×5 and $(B^T A)C + B^T$ is defined, then the size of C is

(a) 2×3

(b) 4×3

(b) 3×3

(d) 3×4

3) If $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 1 & 0 \end{pmatrix}$, then

(a) $AA^T = \begin{pmatrix} 14 & 6 \\ 6 & 17 \end{pmatrix}$

(b) $AA^T = \begin{pmatrix} 14 & 6 \\ 7 & 10 \end{pmatrix}$

(c) $AA^T = \begin{pmatrix} 14 & 6 \\ 6 & 10 \end{pmatrix}$

(d) $AA^T = \begin{pmatrix} 11 & 6 \\ 7 & 10 \end{pmatrix}$

4) Under which condition about a and b so that the following system is consistent:

$$x + 3y = a$$

$$4x + 12y = b$$

(a) $b = -4a$

(b) $b = 5a$

(c) $b = 4a$

(d) $b = -5a$

5) If A is a matrix of size 2×2 and $\det(A) = 2$, then $\det(3A A^T A^{-1}) =$

(a) 36

(b) 18

(c) 32

(d) 16

6) Write a condition about a so that the matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & a & a \\ a & 2 & 0 \end{pmatrix}$ is invertible

(a) $a \notin \{-2, 0\}$

(b) $a \neq -1$

(c) $a \neq 2$

(d) $a \notin \{0, -1\}$

7) If $(2A^T)^{-1} = \begin{pmatrix} -2 & 1 \\ -5 & 3 \end{pmatrix}$, then $A =$

(a) $\begin{pmatrix} 3/2 & -5/2 \\ 1/2 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 3/2 & -5/2 \\ -1/2 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 3/2 & 5/2 \\ -1/2 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} -3/2 & -5/2 \\ 1/2 & 1 \end{pmatrix}$

8) Write a condition about a so that the following system has a unique solution

$$2x + y = 1$$

$$ax - 2y = 3$$

(a) $a \neq 4$

(b) $a \neq -1$

(c) $a \neq 1$

(d) $a \neq -4$

(Show your work)

QUESTION 2 (5 marks)

a) Prove that $H = \left\{ \begin{bmatrix} a & a+b \\ a-b & 0 \end{bmatrix} : a, b \in \mathbb{R} \right\}$ is a subspace of M_{22} .

Solution: First, note that $H \neq \emptyset$ since the matrix 0 belongs to H .

Now, let k be a real number, and let $A = \begin{bmatrix} a & a+b \\ a-b & 0 \end{bmatrix}, B = \begin{bmatrix} a' & a'+b' \\ a'-b' & 0 \end{bmatrix}$ be two matrices of H .

Then

$$\begin{aligned} A + B &= \begin{bmatrix} a & a+b \\ a-b & 0 \end{bmatrix} + \begin{bmatrix} a' & a'+b' \\ a'-b' & 0 \end{bmatrix} = \begin{bmatrix} a+a' & (a+b)+(a'+b') \\ (a-b)+(a'-b') & 0 \end{bmatrix} \\ &= \begin{bmatrix} a+a' & (a+a')+(b+b') \\ (a+a')-(b+b') & 0 \end{bmatrix} \in H. \end{aligned}$$

And

$$kA = k \begin{bmatrix} a & a+b \\ a-b & 0 \end{bmatrix} = \begin{bmatrix} ka & k(a+b) \\ k(a-b) & 0 \end{bmatrix} = \begin{bmatrix} ka & ka+kb \\ ka-kb & 0 \end{bmatrix} \in H.$$

So $A + B$ and kA belong to H . Hence, H is a subspace of M_{22} .

QUESTION 3 (7 marks)

Determine the values of a for which the following system is consistent or inconsistent.

$$x + y + a z = 1$$

$$x + 2y + a z = 2$$

$$x + 2y + 3z = 3$$

Solution:

The augmented matrix is

$$\begin{bmatrix} 1 & 1 & a & 1 \\ 1 & 2 & a & 2 \\ 1 & 2 & 3 & 3 \end{bmatrix}$$

$$R2 \rightarrow R2 - R1, \quad R3 \rightarrow R3 - R1$$

$$\begin{bmatrix} 1 & 1 & a & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 3-a & 2 \end{bmatrix}$$

$$R3 \rightarrow R3 - R2$$

$$\begin{bmatrix} 1 & 1 & a & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 3-a & 1 \end{bmatrix}$$

1) Case 1: $a \neq 3$.

$$R3 \rightarrow 1/(3-a) R3$$

$$\begin{bmatrix} 1 & 1 & a & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1/(3-a) \end{bmatrix}$$

A unique solution $(-a/(a-3), 1, 1/(3-a))$.

2) Case 2: $a = 3$.

$$\begin{bmatrix} 1 & 1 & 3 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

There is no solution because of the contradiction $0 = 1$.